

Allen Sun

Mechanical design engineer, Pittsburgh, PA

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Experience

Curtiss-Wright, Electro-Mechanical Division

Jun 2025 - present

Mechanical Design Engineer since Mar 2026, Pittsburgh, PA

I design and analyze canned induction motor assemblies for marine propulsion. That runs from stator geometry, windings and bearings up to how the air gap and rotor bar choices drive torque, slip and copper loss. I also disposition nonconforming hardware with Solid Edge, FEA and tolerance stack-ups, and I make the GD&T drawings, BOMs and reports that go with these parts.

Ergami Endoscopy

Jan - May 2025

R&D / Robotics Engineering Intern, New York, NY

I designed and built the weld rig behind the TPU tube process for a soft robotic colonoscope insert, using machined aluminum on press-fit brass rods with 3D-printed parts to get consistent heat welds. I also worked on the insert's locomotion concept, from anchor design and actuator selection to motion testing.

Relavo Medical

Sep - Dec 2024

Mechanical Engineering Intern, New York, NY

I worked on a device that disinfects peritoneal dialysis tube connections. I reviewed its design control traceability matrix, wrote verification and validation test protocols, and designed SolidWorks test fixtures for its torque and robustness tests. I also made the GD&T drawings for its injection-molded parts.

Terrament

Sep 2023 - Sep 2024

Mechanical Engineering Intern, New York, NY

I built and wired the 1/24-scale geared drive prototype that Terrament showed at Newlab's New Climate Futures in September 2023. I also ran load calculations, gear-ratio analyses and structural feasibility studies in Fusion 360 for the gravity storage system and its linear harmonic drive redesign. Along the way I printed prototypes around off-the-shelf industrial parts and used FEA to check stress, fatigue life and material choices.

Projects

Micro-vibration canceller

Senior design, NYU, Sep 2024 - May 2025

Our team of four built a bolt-on vibration canceller around a voice coil motor and a piezo sensor, and I machined every part of it along with the test rig. Open-loop tests at 54 Hz cut beam vibration 75 to 90% for 40 to 50 ms at a time, and the faculty voted it Best Senior Design Project for the NYU ME class of 2025.

Robotic arm

Advanced CAD, NYU, Fall 2024

I designed the hand, palm, wrist and forearm of a human-scale arm in Fusion 360, and the wrist held a 1000 N load in SolidWorks Simulation (modeled in steel).

Education

NYU Tandon School of Engineering, B.S. Mechanical Engineering. GPA 3.6/4.0

2025

Skills

Design	Mechanical and product design, DFM, GD&T (ASME Y14.5), fixture design, rapid prototyping, 3D printing, materials selection
Analysis and test	FEA, CFD, tolerance stack-ups, test planning, verification and validation, control systems, system identification, data analysis
Shop	Manual mill and lathe, laser marking, welding
Software	SolidWorks, Solid Edge, Fusion 360, ANSYS (Mechanical, Fluent), MATLAB, Simulink, Python, Arduino, SQL, Power BI
Process	Lean Six Sigma (Green Belt trained), root cause analysis, CAPA